Adapting to Natural Disasters through Better Information: Evidence from the Home Seller Disclosure Requirement

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Abstract

Despite intensifying climate change, US population exposure to flood risk remains high, amplifying the economic toll of floods. While flood risk information policies are gaining traction, it is unclear whether they reduce physical risk exposure or primarily lower financial risk by prompting insurance purchases. Exploiting two quasi-experimental variations in Home Seller Disclosure Requirements, I find that the policy substantially reduces population in high-risk areas by diverting in-migration but has little effect on flood insurance take-up. These results suggest that easing information frictions can spur voluntary climate adaptation in housing market.

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1 Introduction

Since 1980, floods in the United States have caused over $1 trillion in damage, and climate scientists predict they will become more frequent and severe in the future (Milly et al. 2002, NOAA 2020). Despite these warnings, many Americans continue to settle in flood-prone areas, amplifying the economic toll of floods (Weinkle et al. 2018, Redfin 2023, Titus 2023). Recently, flood risk information policies have gained traction as a strategy to reduce flood damage, but their effectiveness remains largely unknown.[1]

Theoretically, a flood-risk information policy can prompt two distinct responses (Ehrlich and Becker 1972). Some individuals may engage in self-protection, actions that directly lower the probability of flood, such as choosing a safer place to live. Others may instead reallocating consumption from the "dry" state to the "flood" state by purchasing flood insurance. Importantly, these strategies have sharply different policy implications: while self-protection reduces physical exposure and thus lowers expected damage at the societal level, insurance primarily redistributes income without mitigating the underlying hazard. This distinction is especially relevant in the flood context, where the National Flood Insurance Program (NFIP)'s heavily subsidized premium structure is known to encourage moral hazard (Peralta and Scott 2024).[2]

This paper exploits quasi-experimental variation from a Home Seller Disclosure Requirement (hereafter, "the disclosure requirement") to assess whether providing flood-risk information prompts households to engage in self-protection or to purchase insurance. I measure self-protection through changes in population distribution across flood-risk zones, and insurance behavior through flood-insurance take-up.[3] The policy mandates that home sellers disclose known property defects to buyers through a standardized form (Lefcoe 2004). Regarding flood risk, a typical question is if a property is located in a Special Flood Hazard Area (SFHA)—an area with elevated risk defined by the official flood map. Given that limited awareness may result from the costs of acquiring and processing information (Kunreuther and Pauly 2004), the disclosure requirement could alleviate the problem by efficiently informing homebuyers about flood risks.

[1] For instance, FEMA has proposed a reform to the National Flood Insurance Program (NFIP) that would make a community's eligibility contingent on mandatory flood risk disclosure (U.S. Department of Homeland Security 2022).

[2] This suggests that, in practice, the two margins function as substitutes despite the theoretical possibility of complementarity. See Section 2 for details.

[3] These are likely the primary margins of adjustment in this setting, though other potential responses are discussed in Section 2.

The disclosure requirement was introduced across 26 contiguous U.S. states between 1992 and 2003. Variation in implementation timing arose from plausibly exogenous state-court rulings on the scope of realtor liability for incomplete disclosure. Coupled with differential disclosure effects for SFHA- versus non-SFHA areas, this staggered rollout facilitates a triple difference research design (Roberts 2006). I also leverage alternative variation stemming from the spatial discontinuity in flood risk information at SFHA borders, which allows me to identify the effect of information while holding true flood risk constant (Noonan et al. 2022). To further control for potentially confounding time-invariant differences at the SFHA border, I use the difference-in-discontinuity approach (Grembi et al. 2016).

I compile multiple datasets to exploit these variations. For population outcomes, I draw on annual NFIP community-level estimates from the US Census Bureau and block-level data from the Decennial Census.[4] In addition to total population counts, the block-level data provide vacancy rates and other demographic characteristics. For flood-insurance outcomes, I use community-level take-up data from Gallagher (2014), extending the series where necessary using Open FEMA data. To measure exposure, I calculate the share of SFHA land in each NFIP community and assign binary SFHA status to Census blocks using FEMA's first-generation digitized flood maps (Q3), which reflect flood risk as of the mid-1990s (FEMA 1996). To trace the legislative history of the disclosure requirement, I rely primarily on *Nexisuni* database.

The empirical analysis yields two main findings. First, the disclosure requirement significantly reduced population in high-risk areas. By analyzing annual community-level population data in a stacked event-study framework with a "binarized" risk exposure variable (Cengiz et al. 2019, Callaway et al. 2024), I find that high- and low-SFHA communities followed nearly identical population trends for a decade before the policy, but relative population in high-risk communities fell by about 4% after five years of policy implementation. Further, I also show that the effect grows monotonically with treatment intensity (i.e., the share of community land within an SFHA). Complementary evidence from a spatial-discontinuity design finds that population in SFHA blocks shrinks by 7% following the policy in comparison to non-SFHA blocks. These changes are driven primarily by diverted in-migration and suppressed new development, rather than out-migration of existing residents, which

[4] A community, as defined by the NFIP, is a local political entity (e.g., village, town, city) that is similar to, but not always aligned with, a US Census Place (Gallagher 2014).

is consistent with the policy mechanism. I also document that these changes led to a higher share of renters and racial minorities in SFHA areas.

Second, I find no evidence that the disclosure policy increased flood insurance take up. Using the same stacked event study framework as in the population analysis, I show that post disclosure take up rates in high-SFHA communities were no greater than in low-SFHA communities. Moreover, the effect size appears unrelated to treatment intensity, which collectively suggest that purchasing insurance was not a primary channel through which homebuyers responded to the policy. Opting for self-protection rather than market insurance may reflect the relative costs and benefits of the two strategies in this context: choosing a safer location is often less costly when buyers are already planning to move, while NFIP coverage offers limited protection, with caps on payouts and exclusions for many indirect losses (Lee et al. 2024).

This paper contributes to two different bodies of literature. First, this paper contributes to research on how housing markets respond to flood risk information. While most existing studies focus on how information shocks affect housing prices—either to test market efficiency (Bin and Landry 2013, Hino and Burke 2021) or to estimate willingness to pay to avoid risk (Pope 2008, Bosker et al. 2019)—I focus instead on how such information reshapes aggregate exposure to flood risk.[5] More broadly, the findings speak to the literature on climate change adaptation. While prior work has primarily emphasized technology as a driver of adaptation (Miao and Popp 2014, Barreca et al. 2016, Burke and Emerick 2016), I highlight how information can help align private decisions with socially desirable outcomes.

Among the first strand of literature, it is worth highlighting how my paper differs from a concurrent study by Fairweather et al. (2024). The authors use a large-scale experiment to show that Redfin, one of the major real estate platform, users are more likely to make offers on safer properties when provided with flood risk information through the website. I complement this study by examining responses to disclosure policies across a broader population, beyond users of a single platform. More importantly, I directly test whether aggregate exposure to flood risk declines in equilibrium. This distinction is critical because individual homebuyers' responses to information may primarily

[5]For example, Hino and Burke (2021) uses flood map updates as the main source of information shock and tests if the housing market efficiently prices flood risk. In contrast, I focus on net population flows and flood insurance demand following an information shock.

affect housing prices and reshuffle who lives where, especially when housing supply is inelastic, rather than reduce total exposure.

Second, I add to the literature on the role of government in shaping household adaptation behaviors (Kousky et al. 2006, 2018, Gregory 2017, Baylis and Boomhower 2022, Peralta and Scott 2024). Perhaps the closest papers conceptually are Baylis and Boomhower (2021) and Ostriker and Russo (2023), which show how building code policies can reduce wildfire damage or flood risk exposure, respectively. A key difference is that the policies studied by these papers directly mandate adaptation, whereas disclosure policies encourage voluntary adaptation such as choosing safer places to live.

The paper proceeds as follows. Section 2 provides background on the Home Seller Disclosure Requirement and the Special Flood Hazard Area. Section 3 details the data sources. Section 4 describes empirical framework and Section 5 show results. Section 6 concludes.

2 Background

Background of policy adoption. Traditionally, homebuyers were expected to practice caution regarding property defects ("*caveat emptor*" or "let the buyer beware" doctrine). However, due to increasing consumer protectionism and public awareness of environmental and health concerns, state courts began holding listing agents accountable for incomplete disclosures (Weinberger 1996, Lefcoe 2004). In response, the National Association of Realtors issued a resolution in 1991 urging state associations to develop and support legislation regarding the statutory disclosure requirement in an effort to deflect potential liability to sellers (Tyszka 1995, Washburn 1995).

Consequently, as Figure 2.1 shows, between 1992 and 2003, 26 states (excluding DC) in the contiguous US adopted a disclosure requirement with an explicit question on flood risk while the remaining 22 states did not implement such a requirement until at least the late 2010s.[6] *(A) Five of the 22 non-disclosure states adopted a home seller disclosure mandate without a question on flood risk. These "placebo" states are useful for checking the robustness of the results.*

Disclosure content. A statutory disclosure requirement mandates that home sellers fill out a standardized form regarding known property conditions and typically deliver it before closing (Stern 2005). Importantly, the disclosure requirement is not exclusively about flood risk. As Appendix Fig-

[6] Appendix Table B.1 reports disclosure timing for disclosed and placebo states.

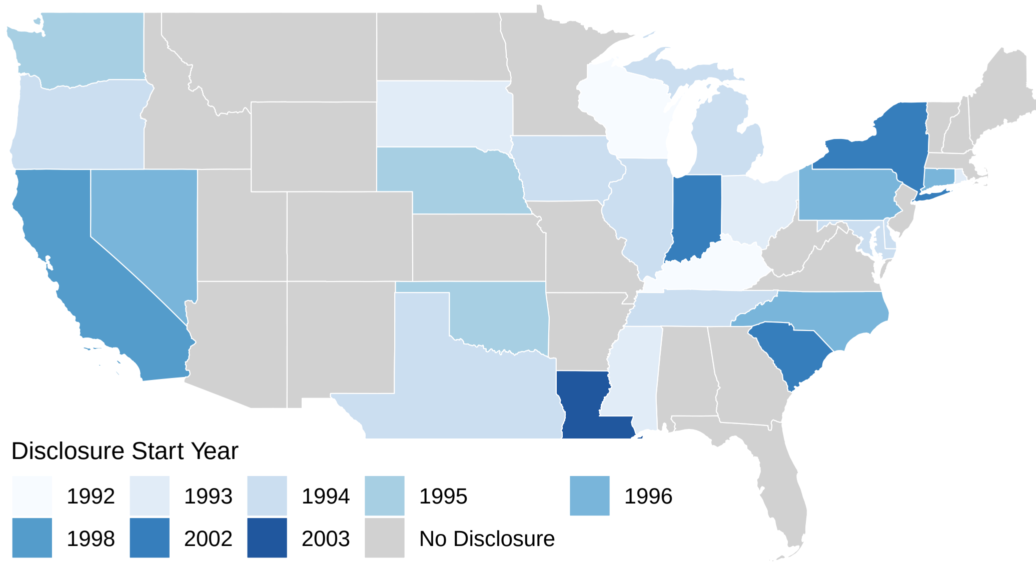


Figure 2.1: The Disclosure Requirement Implementation over Time

ure B.1 illustrates, a typical form covers a wide range of property conditions including structural issues (e.g., problems with walls) and surroundings (e.g., flood risks).[7] This implies that the policy adoption decision is likely to be uncorrelated with underlying flood characteristics or flood policies. Indeed, Appendix Figures B.2 show little correlation between the timing of disclosure requirements and (a) flood risk levels (share of their area in the SFHA), (b) flood insurance take up, (c) flood insurance damage from the NFIP adjuster's report, or (d) other local flood policies (Community Rating System participation).[8]

The exact language of disclosure on flood risk varies slightly from state to state, but some combination of the following three questions usually appears: whether a property is in the SFHA; whether a property has flood damage history; and whether a property has flood insurance.[9] Because properties on the SFHA are more susceptible to flooding, answers to these questions are highly correlated: flood insurance policy and claims data show that 71% (75%) of the claims (flood insurance policies) are from properties in the SFHA. Thus, irrespective of the language, the disclosure requirement is likely

[7]Since the disclosure delivers a bundle of information, discerning treatment mechanism can be challenging if there is a positive correlation between flood risk and other property defects. In Appendix Table B.2, I demonstrate that properties near SFHAs are notably newer compared to those farther away from SFHAs. As property defects typically emerge over time, this table suggests that the disclosure policy's impact likely stems from flood risk information rather than other property defects.

[8]Data sources for Panels (a) and (b) are described in Section 3. For (c) and (d), I use FEMA's official individual claims records (https://www.fema.gov/openfema-data-page/fima-nfip-redacted-claims-v2) and FEMA's NFIP community status book data (https://www.fema.gov/openfema-data-page/nfip-community-status-book-v1), respectively.

[9]As of 2021, 5 states ask just the first question about the SFHA status, 15 states ask about SFHA status and past flood experience, and 4 states ask all three questions. MI and TN ask about the latter two only.

to raise homebuyers' flood risk awareness for properties in SFHAs relative to those outside.

It is worth highlighting that while the disclosure policy helps determine whether a given property is at high flood risk, it provides no information on (1) flood zone boundaries or (2) the flood risk status of other properties on the market. As a result, it is likely difficult for homebuyers to engage in strategic avoidance behavior, such as selectively purchasing properties just outside flood zones, based solely on the information provided by the disclosure.

Flood Map and Special Flood Hazard Area (SFHA). Three facts about SFHA, an area designated by an official flood map for potential inundation by a 100-year flood, are worth highlighting. First, the SFHA boundary is determined by comparing water surface and ground elevations under a 100-year flood scenario (FEMA 2005). This gives rise to the spatial discontinuity design because the disclosure treats flood risk as binary, while actual flood risk varies more gradually across boundaries (Noonan et al. 2022). Second, as shown in Appendix Figure B.4, most communities have only a small fraction of area in SFHA, which suggests that potential treatment spillovers into non-SFHA areas are likely minimal (Busso et al. 2013, Alves et al. 2024). Third, SFHA areas change occasionally due to flood maps updates, although such instances were relatively infrequent prior to 2008 (see Appendix Figure A.1). Such a map update is a source of information shock (Gibson and Mullins 2020, Bakkensen and Ma 2020, Hino and Burke 2021, Weill 2023), which may confound the disclosure effect. In my empirical analysis, I test the robustness of my results against potential spillover effects and changes in flood maps.

Margins of adjustments. Upon learning about flood risk, homebuyers can choose from a menu of strategies. For self-protection, options include selecting a safer location at the outset or purchasing a risky property and later elevating it. In practice, retrofitting is unlikely to be common: buyers can more easily choose an alternative property than incur the large costs of elevation. Median FEMA-assisted elevation projects between 2008 and 2013 cost roughly $166,000 (National Research Council 2015) and entail lengthy permitting and construction periods, during which the property is often unusable.[10]

For insurance, buyers may rely on market insurance or self-insurance (i.e., savings). Historically,

[10]While Ostriker and Russo (2023) finds that floodplain regulations reduce damage largely via elevation, those policies applied to new construction rather than retrofits of existing homes.

flood coverage in the US has been almost entirely provided by the NFIP at heavily subsidized rates: the average subsidies for high-risk properties are around 30% (Wagner 2022). Private market share, though growing, remains only about 3–4% as of 2023 (Kousky et al. 2023).

Given these patterns, my analysis focuses on choosing a safer location as the primary measure of self-protection and NFIP take-up as the measure of insurance behavior. While the discussion so far implicitly treats self-protection and market insurance as substitutes, Ehrlich and Becker (1972) notes that the two can be complements when reduced risk is rewarded with lower premiums. This, however, is unlikely in the NFIP context because premiums are both heavily subsidized and too coarse to account for self-protection efforts (Kousky 2019). Consistent with this, Wagner (2022) finds that substitution between these margins is common in the NFIP setting and Peralta and Scott (2024) further shows that the NFIP encourages households to engage in moral hazard (i.e., residing in more risky areas).

3 Data

Data sources. To determine SFHA status of a given geographic unit, I use the Q3 map—the first generation of a digitized flood map—that reflects flood risk as of the mid-1990s (FEMA 1996). I spatially merge the Q3 flood maps with NFIP community and Census block boundary maps to calculate the share of SFHA land within each NFIP community and to classify Census blocks as either SFHA or non-SFHA. To track changes in flood map over time, I draw on the Community Map History table from FEMA's Flood Insurance Study (FIS) reports.[11]

Annual data on population and flood insurance take-up at the NFIP community level over the 1982—2002 period come from Gallagher (2014). Since NFIP communities closely align with Census Places, population figures are drawn from the Census Bureau's annual population estimates. Following Gallagher (2014), flood insurance take-up is measured as the number of active flood insurance policies per capita, partly reflecting the lack of annual community-level data on the number of insurable structures (homes and businesses).

[11] Practically, I extract the information from the National Flood Hazard Layer (NFHL). The Community Map History table serves as a credible source for tracking map revisions; for example, flood map panel notes, "For community map revision history, refer to the Community Map History table located in the FIS report for this jurisdiction." An alternative source is FEMA's Compendium of Map Updates (Hino and Burke 2021), which documents all changes, including both revisions and initial mappings, to the flood maps. However, I prefer the NFHL as my focus is on identifying revisions to the Q3 map. Further details and validation of the NFHL map revision data are in Appendix A.

Table 3.1: Summary Statistics for Key Variables

Variables	Min	Q25	Median	Mean	Q75	Max	N
Panel A: Community Level Variables							
Population	34	1,000	3,799	17,983	18,863	537,150	27,113
Flood Insurance Take Up	0	0	0.001	0.053	0.007	5.08	27,113
(%) In SFHA	0	0	0	0.189	0.009	1	27,113
Panel B: Block Level Variables							
Population	0	0	13	37.5	44	9,888	2,723,580
(%) Vacant	0	0	0	0.095	0.111	1	2,454,687
(%) Senior (65+)	0	0.056	0.182	0.218	0.333	1	5,145,303
(%) Hispanic Origin	0	0	0	0.079	0.065	1	3,224,921
(%) Owner Occupied	0	0.545	0.833	0.72	1	1	4,202,223
In SFHA	0	0	0	0.121	0	1	2,723,580

I complement community level population data with census-block-level Decennial Censuses. In addition to leveraging total population counts, I compute the share of vacant units among all housing units.[12] I also calculate and use the share of renter-occupied units, as well as the share of household heads who are 65 or older (seniors) or of Hispanic origin, among all occupied units. To account for changing block boundaries and resulting one-to-many matches across different Decennial Census years, I calculate the weighted sum of count variables using interpolation weights from the NHGIS block-to-block crosswalk (Manson et al. 2022).[13]

Finally, the primary data source to track the disclosure requirement legislative history is the *Nexisuni* database. I cross-validate this database with prior studies on the disclosure requirement (Washburn 1995, Pancak et al. 1996, Lefcoe 2004) and reports from the National Association of Realtors (National Association of Realtors 2019).

Descriptive statistics. Table 3.1 reports summary statistics for the main sample used in the empirical analysis in Section 5. Panel A presents community-level variables, focusing on communities where more than 70% (or less than 1%) of land area falls within the SFHA based on the "(%) In SFHA" variable. Note, in rare cases—0.9% of observations—flood insurance take up (i.e., flood insurance counts per capita) exceeds one, as seen in the maximum value in the table, because some

[12] A property is considered vacant if no one is residing in the unit at the time of enumeration unless its occupants are only temporarily absent (US Census Bureau 2000).

[13] For instance, block G06000104003003006 in 2000 is matched to five different blocks in 2010 ending in 3010, 3011, 3017, 3020, and 3028. More generally, 26% of blocks from 2000 and 41% from 1990 are matched to multiple blocks in 2010. Interpolation weights represent the expected proportion of the source block's counts (e.g., population or housing units) located in each target block (Manson et al. 2022).

communities have many structures but relatively few residents.[14]

Panel B shows block-level variables. A two points are worth discussing. First, because an optimal bandwidth for spatial discontinuity design is outcome variable specific, the number of observations used for analysis vary. Second, 33% of observations for the block population are zero. In addition, these variables also exhibit substantial skewness (long and thin right tails), as the difference between median and mean values suggests. To account for this, I follow Chen and Roth (2022) and estimate extensive and intensive margin effects separately when I leverage block level data. Also, in Appendix A, I show that the frequency of zeros in my data is consistent with external sources.

4 Empirical Framework

To assess the impact of the disclosure policy on home buyer responses, I begin by analyzing annual community-level data using an event study design in equation (1).

$$Y_{msztd} = \sum_{k=-10}^{4} \beta^k H_m D_{sd} \mathbb{1}(t - t_s^* = k) + \theta_{std} + \lambda_{zdt} + \psi_{szd} + \epsilon_{msztd} \tag{1}$$

Y_{msztd} denotes the log population or per capita flood insurance take up for community m in state s in flood risk status z, in year t, and stack d. H_m is a binary indicator for whether community m belongs to group z, where $z \in \{\text{high-SFHA}, \text{low-SFHA}\}$. $\mathbb{1}(t - t_s^* = k)$ is an event time indicator equal to 1 if year t is k years from the policy change year t_s^* in state s, for $k \in \{-10, -9, \cdots, 4\}$. D_{sd} is an indicator for whether state s belongs to the disclosure group in stack d. Other terms in a standard triple difference model are subsumed by fixed effects. The coefficient β^k captures the effect of the disclosure policy in high-risk communities, relative to low-risk communities, in disclosed states relative to not-yet-disclosed states, at event time k.

A three points are worth noting. First, equation (1) takes a stacked triple difference approach to overcome potential bias from conventional two-way fixed effect models under treatment heterogeneity (Cengiz et al. 2019, Goodman-Bacon 2021). Importantly, I use late adopting states as a control group in equation (1) because as Appendix Table B.3 shows, (1) the 22 never-adopted states are different in baseline demographic, economic, and political characteristics from the 26 ever-adopted

[14] There are 8 unique communities with flood insurance per capita exceeding 1. All are summer destinations, which have small year-round populations.

states but (2) such a difference does not appear in the early—14 states that have implemented the policy by 1994—vs. late—12 states implemented after 1994—adopting states comparison.[15] Moreover, consistent with these baseline differences, when I use never-adopting states as controls, the event-study estimates in Section 5.1 exhibit pronounced pre-trends. Thus, each stack d consists of communities in the treated states, which adopted the disclosure policy in year t^*, and communities in the control states, which adopted the policy in $\tilde{t} > t^*$.[16] Each d covers a window of 10 years before and 5 years after t^* and sample composition during this period remains stable.[17] I drop observations from the control states for $t >= \tilde{t}$ because they are no longer "not-yet-treated".

Second, because disclosure intensity is defined as the fraction of land area within the SFHA, it is inherently a continuous measure. Recent work shows that difference-in-differences with a continuous treatment can be biased when treatment effects vary with intensity (Callaway et al. 2024). To address this problem, I "binarize" the treatment intensity variable: that is, I set $H_m = 0$ for communities with SFHA share less than 1% (effectively no SFHA) and $H_m = 1$ for those with SFHA share above 70% (roughly the bottom 10% vs. top 3% of the distribution), dropping all intermediate cases. I also report sensitivity to the choice by relaxing the high-SFHA cutoff to 50% and 30%.

Third, Because equation (1) leverages staggered, state-by-year adoption and includes state-year-stack (θ_{std}) and high flood risk group-year-stack (λ_{zdt}) fixed effects, the specification controls for any concurrent policies or shocks that: (1) affect all communities within a state equally, (2) affect high-SFHA communities uniformly across states—such as federal-level flood policies, or (3) affect high-SFHA areas within a state, but whose timing is not systematically related to the timing of disclosure adoption.

However, equation (1) also has a few limitations. First, because communities do not perfectly align with Census Places (Gallagher 2014), a large number of communities are excluded from the estimation. Second, since location responses to flood risk could be often local (Elliott and Wang 2023), β^k—which cannot capture intra-community relocation—may significantly underestimate the

[15] Deshpande and Li (2019) also exploit the timing of treatment because eventual treatment status was predictable based on covariates. Roth and Sant'Anna (2023) also states that in non-experimental contexts, the quasi-randomness in identifying variation may be more plausible when never-treated units are excluded.

[16] Stack refers to data that is created for a specific treatment year (or a cohort year). A state can belong to both treatment or control groups depending on the stack. For instance, PA and CT, which changed their policy in 1996 are in the "treatment group" in a stack for $t^* = 1996$. The two states belong to the "control group" when $t^* < 1996$.

[17] Because the policy adoption is complete by 2003, "not-yet-treated" controls only exist through 2002. Thus, the 2002 cohort has just one post year with corresponding not-yet-treated controls (2002), and the 2003 cohort has none. Accordingly, I include the 2002 and 2003 cohorts only as control units.

disclosure policy's effect on population distribution. Third, demographic data are unavailable at the community-by-year level.

Thus, I complement equation (1) by analyzing census block-level population data, leveraging a spatial discontinuity in flood risk information from the disclosure at the SFHA border.[18] Importantly, because other policies such as flood insurance requirements also change at the border, I employ a difference-in-discontinuity approach following Grembi et al. (2016) and Gottlieb et al. (2022), which controls for time-invariant cross-sectional differences between non-SFHA and SFHA areas.[19]

$$\begin{aligned} Y_{bst} =& \delta_0 + \delta_1 X_{bs} + \delta_2 D_{bs} + \delta_3 X_{bs} * D_{bs} + \\ & T_{st}[\delta_4 + \delta_5 X_{bs} + \delta_6 D_{bs} + \delta_7 X_{bs} * D_{bs}] + \epsilon_{bst} \end{aligned} \tag{2}$$

In equation (2), Y_{bst} represents population outcomes in block b in state s in time t. X_{bs} is the distance between a block border and the closest SFHA border in meters (negative if in a non-SFHA area), which is approximated by taking the difference of (1) the distance between block centroids and the closest SFHA border and (2) a block diameter. $D_{bs} = 1$ if $X_{bs} > 0$ is a high risk group (i.e., SFHA) indicator variable, and $T_{st} = 1$ if $t > t^*_s$ is a post disclosure-period indicator variable. δ_6 captures the impact of the disclosure policy for blocks located in close proximity to the border.

To estimate δ_6, I first estimate the optimal bandwidth for each outcome variable. Then, I estimate equation (2) using blocks within the optimal bandwidth (Calonico et al. 2014, Cattaneo et al. 2019).[20] For states that implemented disclosure policies between 1990-1999 (2000-2009), the 1990 (2000) decennial census is used to estimate δ_0 to δ_3 whereas 2000 and 2010 (2010 and 2020) decennial censuses are used to estimate δ_4 to δ_7. Also, I remove 17% of blocks that contain SFHA borders from the analysis because X_{bs} may not be well defined for them.

(B) Notably, as Grembi et al. (2016) points out, equation (2) differs from studies that apply a difference-in-differences approach within a regression discontinuity (RD) style sample to improve comparability between treated and control units (Greenstone and Gallagher 2008, Lemieux and Milligan

[18]While Noonan et al. (2022) shows that (true) flood risk changes continuously at the border in Texas, true risk may change more abruptly in other states within my sample. However, because flood risk, which is largely a function of land contour, is likely to remain stable at least in a relatively short term, such differences will also be controlled by the difference-in-discontinuity approach.

[19]There might be a concern for time-varying policy changes at the border as well. However, the results from equation (1) mitigate these concerns, and I present additional robustness checks in Section 5.1.

[20]I estimate the mean squared error optimal bandwidth for 2000 and 2010 and take the average of them following Grembi et al. (2016). I ignore 1990 and 2020 because these years have only a subset of the states in the sample.

2008, Pettersson-Lidbom 2012), as their identification still relies on the parallel trends assumption.[21] *In contrast, in equation (2), identification rests on the assumption that there are no other important time-varying differences at the discontinuity.*[22]

Throughout Section 5, standard errors are clustered at state—the level of disclosure treatment.

5 Results

5.1 The Effect of the Disclosure on Population

Figure 5.1 (a) reports $\hat{\beta^k}$ from equation (1), where $H_m = 1$ for communities with more than 70% of their area located within the SFHA. Over the ten years preceding the disclosure policy, population trends in high-SFHA communities closely tracked those in low-SFHA communities (defined here as having less than 1% of their area in the SFHA). In the first two years following policy implementation (event times 0 and 1), there is little evidence of population change in high-SFHA communities.[23] Starting in year 3, however, these communities begin to experience population declines, with the effect reaching nearly 4% five years after implementation. The later-period effects are statistically significant at the 90% level, whereas the immediate post-policy years are not.

Panel (b) shows how results change when high-SFHA communities are instead defined as those with SFHA shares above 50% or 30% (rather than 70%), each compared to communities with less than 1% SFHA area. The figure overlays the estimates from Panel (a) with those from alternative thresholds. The coefficients display a clear monotonic attenuation: as the threshold is lowered (i.e., treatment intensity becomes lower on average), the estimated population decline becomes smaller and, in some cases, even turns slightly positive—though imprecisely estimated (Appendix Table **??**).[24]

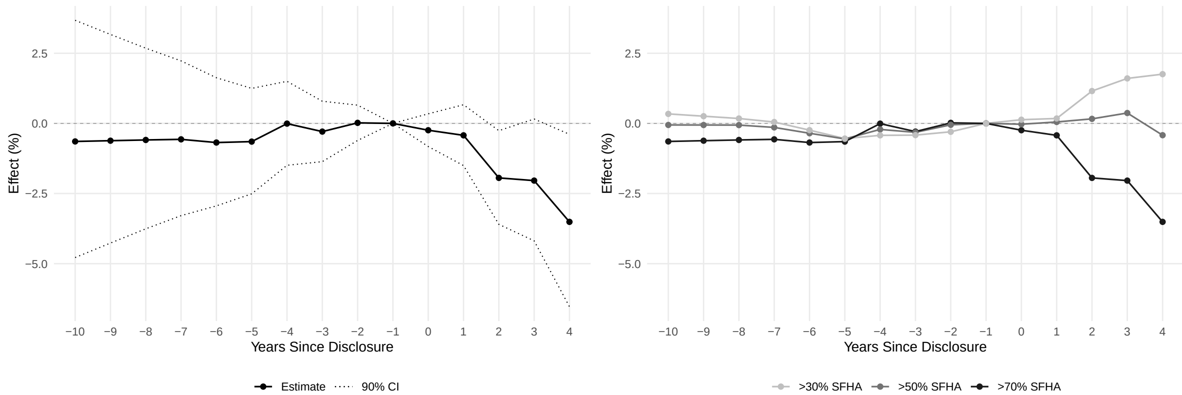
In Appendix Figure B.5 (a), I present an event study plot that include all 22 never-adopted states

[21] Analyzing the decennial data using a version of equation (1) has at least two critical challenges. First, there is only a single post-treatment period because, by 2010, there are no "not-yet-treated" states remaining. Second, there is only one pre-treatment period unless I forgo about 80% of observations due to the 1980 Census covering only urban or metropolitan areas. Note that these issues do not arise in my analyses using annual data.

[22] Thus, the discussion on control group selection for equation (1)—between never-treated and not-yet-treated units—is irrelevant for equation (2) as non-SFHA blocks near the SFHA border, which are similar to SFHA blocks afer netting out time-invariant differences, are effectively serving as a control group in equation (2).

[23] *(C) One potential explanation is that, with housing supply fixed in the short run, the policy's initial effects may have occurred mostly through price adjustments and reshuffling rather than shifts in population.*

[24] I plan to add the table later.



(a) Event Study Estimates

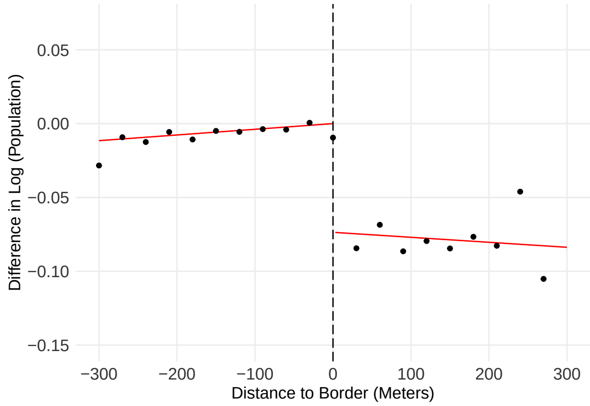
(b) Alternative High-Risk Definitions

Figure 5.1: The Effect of Disclosure on Community Level Population. Panel (a) reports the impact of the disclosure policy on community-level population, namely β^k in equation (1), for varying thresholds of $H_m = 1$. Panel (b) plots the estimated coefficients from an event-study specification of equation (1), where high-SFHA communities are defined as those with more than 70% of their area in the SFHA ($H_m = 1$). Error bars represent the 90% confidence interval.

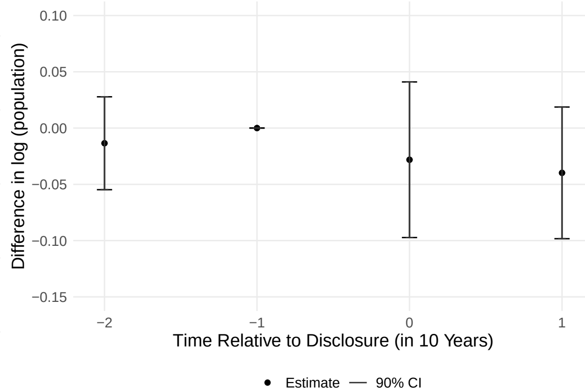
as a control group. In contrast to Figure 5.1 (a), Appendix Figure B.5 (a) has a noticeable pre-trend, which implies that the never-adopted states are unlikely to satisfy the parallel trend assumption. This is also consistent with differences in baseline characteristics as discussed in Section 4. Appendix Figure B.5 (b), I show that the result is robust to the exclusion of communities that had experienced flood map updates during the sample period.

Take together, Figure 5.1 suggest that homebuyers tend to choose to reside in lower-risk communities rather than high-risk ones after receiving flood risk information, thereby reducing flood risk exposure. However, Figure 5.1 (a) may underestimate the policy impact on population since homebuyers can substantially lower their flood risk without necessarily making a large location adjustment. For instance, 58% (74%) of recipients in FEMA's buyout program for flood-prone properties relocate within a 10-mile (20-mile) drive of their original homes while still reducing their flood risk score by over 60% (Elliott and Wang 2023).

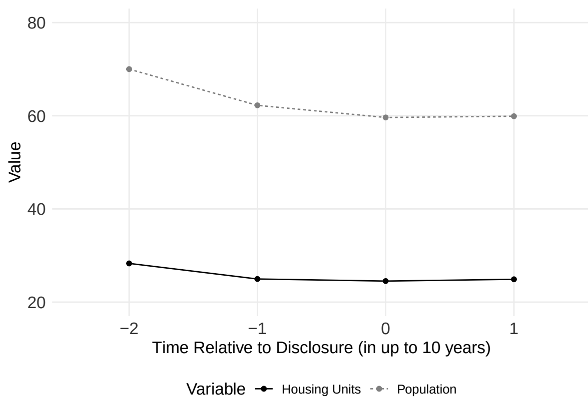
Motivated by this, I exploit census block-level data to explore more granular location responses, with a caveat that δ_6 in equation (2) represents the local average treatment effect near the SFHA boundary. While the magnitude may differ in other areas unless homogeneity in treatment effect is assumed, it should be noted that the area near SFHA boundary still captures a substantial portion of flood-prone locations, given that only 8% of a typical community land falls within SFHA (Appendix Figure B.4).



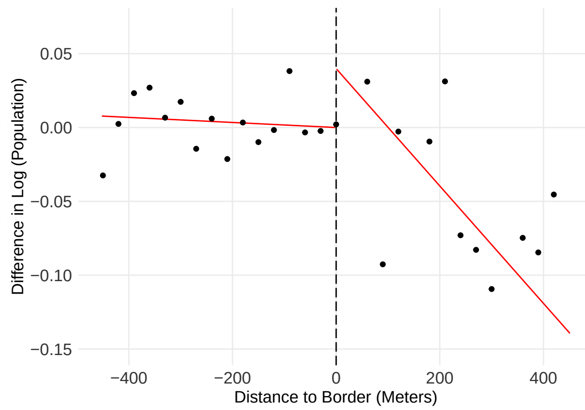
(a) Change in Spatial RD Estimates (Disclosed States)



(b) Spatial RD Estimates Over Time



(c) Raw Trends for SFHA Blocks



(d) Change in Spatial RD Estimates (Placebo States)

Figure 5.2: The Effect of Disclosure on Population. Panel (a) illustrates difference-in-discontinuity estimates for the disclosed states. The dependent variable is the change in log of population after the disclosure. The running variable is the distance between a census block and the nearest SFHA border. The discontinuity at the threshold (dashed vertical line) represents δ_6 in equation (2). Panel (b) plots the disclosure effect over time relative to the disclosure policy change timing. Panel (c) plots population and housing unit trends (in absolute terms) over time for SFHA blocks. For Panels (b) and (c), I limit the sample to blocks that are observed in at least four decennial censuses. Panel (d) repeats Panel (a) for the placebo states

Figure 5.2 Panel (a) illustrates the impact of disclosure policy on log population. The horizontal axis is X in equation (2) and blocks with $D = 1$ are presented on the right-hand side of the border. The vertical axis is the difference in log population between pre and post treatment periods. The solid lines is the regression fit and points reflect the difference in log population for each distance bin. δ_2 from equation (2) is normalized to 0 to enhance readability.

The figure indicates that there is a 7 percent drop in the population for SFHA blocks relative to non-SFHA blocks at the SFHA boundary after the disclosure requirement. The tight fit of the regression line to the scatter plot suggests that the choice of functional form for the running variable likely has minimal impact on the estimates.

(D) In Figure 5.2 (b), I plot coefficients correspond to δ_2 from the equation $Y_{bst} = \delta_0 + \delta_1 X_{bs} + \delta_2 D_{bs} + \delta_3 X_{bs} D_{bs}$, estimated separately for each time period relative to the disclosure policy change, and normalized to δ_2 at relative time -1.[25] *The analysis focuses on blocks that appear in at least four decennial censuses, which implies that 80% of observations excluded due to the 1980 census covering only urban or metropolitan areas. Although the results are underpowered due to this sample restriction, Panel (b) indicates that there was no population difference between SFHA and non-SFHA areas before the disclosure, but a 3-4% relative decline in SFHA area population after the disclosure.*

Interestingly, the raw population and housing units for an average SFHA block in Panel (c) remains constant even after the disclosure, suggesting that the population adjustments stem from diverted in-migration and suppressed development, rather than active out-migration. This is consistent with the policy's role in informing prospective buyers rather than existing homeowners.
Table 5.1 presents additional results. Column (1) reports the intensive margin effect, restricting the sample to blocks with a nonzero population, showing a -7% reduction in SFHA population. This estimate corresponds to Panel (a) of Figure 5.2. Column (2) reports the extensive margin effect—the disclosure reduces the probability of having any population in an SFHA block by 0.01 relative to a non-SFHA block (or 1.5 percent of the baseline value of 0.68).[26]

Consistent with the population effects, column (3) shows that the disclosure increases the vacancy rate for an SFHA block from 0.095 to 0.109. In line with this higher vacancy, column (4) reports a roughly 0.7 percentage point increase (or 2.5% increase from the baseline) in the share of renter-occupied units among occupied housing. Together, columns (3)–(4) suggest that, after the disclosure, selling properties in SFHA areas becomes more difficult (or takes longer), leaving a larger fraction vacant at any given time, with some of those units transitioning to rental.[27] More broadly, columns (1)–(4) align with prior research documenting population declines following negative environmental shocks (Banzhaf and Walsh 2008, Boustan et al. 2012, Hornbeck and Naidu 2014).

In columns (5)-(6), I further examine how demographic composition changes after the disclosure policy and find that the fraction of minority population, in particular, those with Hispanic Origin, in-

[25] Relative time indicators are defined as -2 (-1) for 19 to 10 (9 to 1) years before the policy change, and 0 (1) for 0 to 9 (10 to 19) years after the policy change.

[26] Note that the number of observations in column (1) is *greater* than in column (2), despite its sample restrictions (i.e., focusing on non-zero population blocks), because the estimated optimal bandwidth is larger for column (1). More generally, the estimated optimal bandwidths in Table 5.1 are outcome variable specific.

[27] Indeed, New Orleans, one of the highest flood-risk areas in the nation, has the highest vacancy rate among the 75 largest MSAs in the US (Fudge and Wellburn 2014).

Table 5.1: The Effect of Discosure on Net Block Population Flow

	Log Population	Prob. of Any Population	Vacancy Rate	Renter Occupancy Rate	Hispanic Occupancy Rate	Senior Occupancy Rate
	(1)	(2)	(3)	(4)	(5)	(6)
SFHA × Post	$-.074^{**}$	$-.011^{***}$	$.014^{***}$	$.007^{**}$	$.008^{**}$	$-.005$
	(.030)	(.003)	(.004)	(.003)	(.003)	(.004)
Avg D.V.		0.675	0.095	0.28	0.079	0.218
Bandwidth	301	138	262	564	380	798
Num. obs.	1915717	1483356	1700002	2952205	2243413	3650471

Note: Columns (1)–(6) are estimated based on equation (2) using the decennial census block-level data. Standard errors are clustered at the state level. $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

creases by about 0.8 percentage points (or 10% increase from the baseline). This pattern aligns with Bakkensen and Ma (2020), who find that households with fewer resources tend to sort into riskier areas after a flood risk information shock. The share of senior residents (age 65 and above) also falls by about 0.5 percentage points, although the estimate is statistically insignificant. A reduction in older residents is plausible given their lower physical capacity to cope with potential flooding.

Robustness checks. Two potential concerns regarding the difference-in-discontinuity results merit attention. First, time-varying confounders may affect the results. Over a 10-year period, many factors could change differentially between SFHA and non-SFHA areas, such as the enforcement of flood insurance purchase requirements. Second, the disclosure requirement may have a spillover effect (i.e., violate the stable unit treatment value assumption (SUTVA)) (Donaldson 2015). For instance, homebuyers who would have chosen properties on SFHAs may instead choose nearby properties in non-SFHAs after the disclosure policy.

Regarding the first issue, it is important to note that the effects in Figure 5.1 are identified from the staggered adoption of disclosure policies across states over multiple years, after controlling for state-specific and high-risk-community-specific annual shocks. *(F) This implies that potential confounders are unlikely to bias the estimates unless they affect specific states and their timing is systematically correlated with the timing of disclosure adoption.* As for the second issue, strategic avoidance based on disclosure information is unlikely, as discussed in Section 2, since the policy only provides risk information for a given property, not broader flood risk patterns.

Additionally, I conduct a series of formal robustness tests to further assess these concerns using the outcome variable of main interest, i.e., log of population. *(E) For the first issue, I start with Figure 5.1 Panel (d), which repeats the same exercise as Panel (a), but using five placebo states that have implemented a version of disclosure requirement that does not provide information on flood risk. Consistent with the interpretation that the policy's impact operates through the disclosure of flood risk information, Panel (d) shows no evidence of a population decline—if anything, an increase in point estimate—following the disclosure implementation.* Second, in Appendix Table B.4 column (1), I estimate equation (2) after removing blocks that had flood map update(s) during the sample period. The idea is that the map update might be a source of time-varying information shock that affects SFHA and non-SFHA areas differentially. I find that the effect size remains very similar. Third, I modify equation (2) and allow time-varying discontinuity at the SFHA border to more broadly control for potential time-varying changes at the border.[28] The estimated coefficient in Appendix Table B.4 column (2) shows the same sign as the main model and if anything, a much larger (in magnitude) effect.

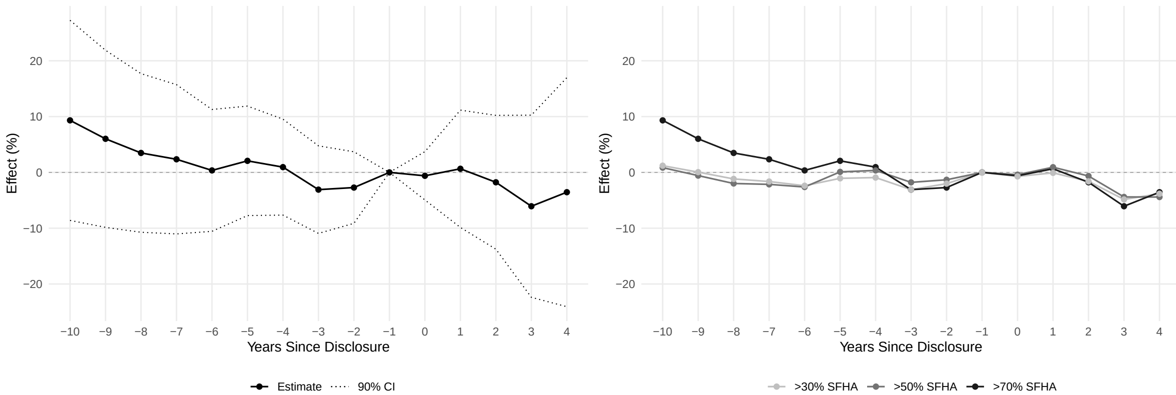
To more formally test potential spillover effects, I repeat my analysis using a doughnut difference-in-discontinuity approach that excludes blocks very close to the border (Kline and Moretti 2014). The idea is that if there is endogenous sorting near the border, the treatment effect may change when those observations are excluded (Cattaneo and Titiunik 2022). Appendix Table B.4 columns (3) and (4) show that the estimates remain nearly identical even if I remove blocks within 20 to 40 meters from the border. Similarly, the policy effect does not diminish even if I expand the bandwidth or use progressively farther away blocks as control units (Appendix Figure B.6 Panels (a) and (b)).[29]

5.2 The Effect of the Disclosure on Flood Insurance Take Up

To examine whether homebuyers respond to flood-risk information by purchasing insurance, I estimate equation (1) using the flood insurance take-up rate, which is defined as per capita NFIP policy

[28] For this, I estimate $Y_{bst} = \delta_0 + \delta_1 X_{bs} + \delta_2 D_{bs} + \delta_3 X_{bs} D_{bs} + \sum_{t=\{1990,2000,2010,2020\}} Q_t[\delta_0^t + \delta_1^t X_{bs} + \delta_2^t D_{bs} + \delta_3^t X_{bs} D_{bs}] + T_{st}[\delta_4 + \delta_5 X_{bs} + \delta_6 D_{bs} + \delta_7 X_{bs} D_{bs}] + \epsilon_{bst}$ where Q_t is an indicator that takes 1 if the time period is in $t = \{1990, 2000, 2010, 2020\}$ and the rest of the notations follow equation (2). The coefficient of interest is δ_6 as before.

[29] Note that in Panel (a), the standard errors remain relatively stable as the bandwidth increases. This is likely due to state-level clustering: because increasing the bandwidth does not increase the number of clusters (states), the additional observations do not meaningfully reduce standard errors. In contrast, when standard errors are calculated without clustering, they decline monotonically with bandwidth. In Panel (b), I estimate equation (2) using control blocks that are within the distance of $(r-1) \times$ *optimal bandwidth* and $r \times$ *optimal bandwidth* for $r \in \{1, 2, 3, 4, 5\}$.



(a) Event Study Estimates

(b) Alternative High-Risk Definitions

Figure 5.3: The Effect of Disclosure on Flood Insurance Take Up. Panel (a) reports the impact of the disclosure policy on flood insurance take up rate, namely β^k in equation (1), for varying thresholds of $H_m = 1$. Panel (b) plots the estimated coefficients from an event-study specification of equation (1), where high-SFHA communities are defined as those with more than 70% of their area in the SFHA ($H_m = 1$). The error bar represents the 90% confidence interval.

counts, as the outcome variable[30] Similar to Figure 5.1 (a), Figure 5.3 (a) shows the differential impact of the disclosure policy for high-risk communities, where $H_m = 1$ for communities with more than 70% of their area in the SFHA, relative to those with less than 1%. The results provide little evidence of increased take-up: aside from event time 1, which shows a small positive impact, all post-disclosure coefficients are near zero or negative.[31]

Figure 5.3(b) overlays event study estimates from alternative models defining high-SFHA communities as having more than 50% or more than 30% of their area in the SFHA, instead of the 70% cutoff. In contrast to the population results, where effect sizes attenuate as lower-intensity observations are included, the post-disclosure point estimates here are essentially identical across all three thresholds. Together with results in Panel (a), this suggests that high-SFHA communities do not seem to respond to the information shock by purchasing flood insurance.

Opting for self-protection instead of market insurance may reflect relative benefits and costs of the two different strategies in this setting: choosing a safer location is often less costly for homebuy-

[30] Ideally, I would observe insurance purchase decisions at the household level. In practice, community-level take-up rates serve as a useful proxy, since the disclosure policy is unlikely to affect incumbent residents' insurance decisions. Because only buyers engaged in a transaction receive the disclosure form, changes in community-level rates are likely driven by the purchase behavior of new entrants.

[31] Note, to express the results in percentage terms, I divide the estimated coefficient by the mean flood insurance take-up rate of high-SFHA communities. I do not log-transform the outcome because over 20% of control communities (with SFHA area below 1%) have zero insurance policies, which would result in substantial loss of observations if logs were used.

ers, as they plan to move regardless and typically consider more than 10 properties before closing (Zumpano et al. 2003). Moreover, the NFIP coverage may provide limited benefits: it does not (1) cover asset losses over $250,000 and (2) compensate for numerous economic costs (e.g., loss of income or use value) beyond asset losses (Lee et al. 2024).

6 Conclusion

Floods are the costliest natural disaster in the US and are expected to become more frequent and severe in the future. Yet the US population exposure to flood risk remains persistently high. Despite growing attention to flood risk disclosure policies, evidence remains limited on whether such measures alter homebuyer behavior in ways that reduce overall exposure to flood risk, as opposed to merely redistributing financial burdens across potential outcomes.

By exploiting plausibly exogenous variations created by a Home Seller Disclosure Requirement, I find that providing more flood risk information to homebuyers reduces the population residing in high-risk areas by 4–7%, with little corresponding change in flood insurance take-up. Given that average annual flood losses for a typical SFHA property are three times greater than those for a non-SFHA property (Gourevitch et al. 2023), and that the compliance costs for this policy are likely moderate at most,[32] the policy likely generates substantial welfare gains. These findings indicate that alleviating information frictions regarding flood risk in the housing market can facilitate voluntary adaptation by helping homebuyers make more informed choices.

(A) in Section 2: Should I keep this or not? I'm worried that it might invite questions like "why not tryinig placebo states for Figure 5.1, for instance?

(B) in Section 4: Do I need to mention this? I initially brought this in because there was referee confusions about what I do with spatial discontinuity design. In particular, they seem to think that I run a diff-in-diffs regression using the RD sample. Also, they complained that in this diff-in-diffs models, I need to use never-treated states as a control but it's not relevant here.

(C) in Section 5.1: Does this statement make sense? Potential pushback?

[32] Moore and Smolen (2000) documents that home sellers spend less than 40 minutes to fill out the disclosure form.

(D) in Section 5.1: Is this helping or hurting (confusing)? This is related to point (b) made earlier, but I said spatial discontinuity design does not rely on event study, but present this event-study style figure. So people might get confused? Maybe I call this “placebo test” in time?

(E) in Section 5.1: Is this helping or hurting? Again, I think there’s a good reason to not to do this exercise with the triple difference design but it can make readers confused? On the other hand, I think it’s a strong robustness check.

(F) in Section 5.1: Does this statement make sense? Potential pushback?

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A Appendix A: Data Validation

Key dependent variables. A prevalence of zeros are consistent with findings from external sources. For block population, Bureau of the Census (1994) reports that a substantial number of blocks in the 1990 Decennial Census have zero population, with state-level proportions ranging from 14% (RI) to 65% (WY), and a median value of 31% (WA). In my sample, the numbers are slightly different at 17% for RI and 26% for WA (WY is a non-disclosure state). A minor discrepancy is not surprising given that blocks not included in the digitized flood map are excluded from the analysis.

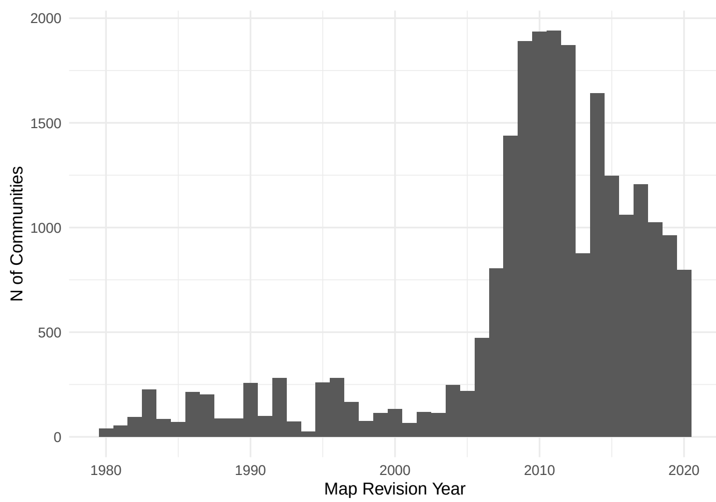

Figure A.1: The Number of Communities with Flood Map Revisions by Year. This figure illustrates the number of communities with flood map revision from 1980 to 2020 using Community Map History from the Flood Insurance Study (FIS) reports.

Flood map revision data. According to the Community Map History from the Flood Insurance Study (FIS) reports, 10.8% of communities in my community-level sample revised their flood maps during the sample period—covering 10 years before and after each state's disclosure policy change. In comparison, in my block-level sample, 48.2% of communities revised their flood maps during the sample period, defined as 1990–2010 for states treated before 2000 and 2000–2020 for states treated after 2000. These block-level windows are chosen to align, as closely as possible given decennial census constraints, with the 10 years before and after each state's policy change.

Such a stark difference is explained by Appendix Figure A.1, which plots the distribution of flood map revision years from the FIS reports. The figure shows a sharp increase in revisions beginning in 2008. States implementing the policy in the 1990s (latest cohort year 1998) have 10-year windows that largely avoid this surge, minimizing its influence on the community-level sample. In contrast, the block-level sample's measurement windows overlap with the post-2008 spike, making them much more exposed to the surge in map updates.

It is also worth noting that FEMA's Compendium of Map Updates (Hino and Burke 2021), which is an alternative source of map updates, seems to document any changes made in a given year. For example, in 2009, communities 371038, 42027C, 21189C, and 17083C are all listed in the compendium as having map updates, but the corresponding FIS reports indicate that these maps were created for the first time. Since my goal is to account for potential changes in flood zones delineated in the Q3 flood map, I focus exclusively on map revisions. The Community Map History data from the FIS reports is therefore more appropriate for this purpose.

B Appendix B: Additional Tables and Figures

Table B.1: Disclosure Policy Adoption by Year

Policy Change Year	Disclosed	Disclosed (w/o Flood)
1992	KY, WI	
1993	MS, OH, RI, SD	
1994	DE, IA, IL, MD, MI, OR, TN, TX	ID, NH
1995	NE, OK, WA	KS
1996	CT, NC, NV, PA	
1998	CA	
1999		ME
2002	IN, NY, SC	
2003	LA	MN

Note:
This table presents the list of states that adopted a disclosure policy by year. The last column includes states that implemented a disclosure policy without a question on flood risk.

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Table B.2: Building Age by the SFHA Status

	N of Houses (< 5Yrs)	N of Houses (> 40Yrs)	(%) Houses (< 5Yrs)	(%) Houses (> 40Yrs)
	(1)	(2)	(3)	(4)
SFHA > 0	119.507***	−254.004***	.057***	−.192***
	(13.437)	(29.864)	(.007)	(.034)
Constant	73.182***	661.449***	.041***	.471***
	(24.349)	(54.139)	(.010)	(.053)
Num. obs.	32391	32391	31490	31490

Note: This table compares the proportion of older and newer housing stocks in census tracts with and without SFHA areas within a tract using the 1990 decennial census data. Standard errors are clustered at the state level. $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

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Table B.3: State Characteristics in 1990 by Disclosure Status

	Ever/Early		Never/Late		Difference	
Variables	Mean	SE	Mean	SE	Mean	P.Value
Panel A: Ever vs. Never States						
Population (millions)	6.57	1.31	3.43	0.651	3.143	0.048
Median Age	33.04	0.204	32.82	0.409	0.22	0.616
(%) White	0.827	0.019	0.879	0.018	-0.053	0.051
(%) BA	0.121	0.005	0.129	0.006	-0.007	0.324
Unemployment Rate	0.06	0.003	0.061	0.002	-0.001	0.773
GDP (billions)	152	34.38	74	14.95	78	0.057
N Housing Units (millions)	2.66	0.506	1.47	0.291	1.187	0.059
(%) Vacancy	0.095	0.005	0.132	0.008	-0.037	0
Democratic Party Vote Share	0.455	0.01	0.425	0.012	0.03	0.06
Flood Insurance Per Capita	0.036	0.014	0.019	0.007	0.017	0.306
Flood Damage (thousands)	48.64	23.66	24.45	6.41	24	0.375
(%) in SFHA	0.156	0.011	0.128	0.013	0.028	0.105
Panel B: Early (Before 1994) vs. Late (After 1994) States						
Population (millions)	5.53	1.29	7.8	2.42	-2.274	0.397
Median Age	33.07	0.286	33	0.302	0.071	0.865
(%) White	0.842	0.026	0.808	0.027	0.034	0.374
(%) BA	0.119	0.006	0.124	0.008	-0.005	0.592
Unemployment Rate	0.061	0.004	0.06	0.004	0.001	0.89
GDP (billions)	119	29.72	191	66	-72	0.306
N Housing Units (millions)	2.25	0.527	3.12	0.917	-0.87	0.402
(%) Vacancy	0.095	0.007	0.096	0.007	-0.001	0.908
Democratic Party Vote Share	0.47	0.013	0.438	0.014	0.031	0.118
Flood Insurance Per Capita	0.037	0.024	0.034	0.013	0.003	0.912
Flood Damage (thousands)	62	43.02	33.53	12.86	28	0.565
(%) in SFHA	0.153	0.01	0.16	0.022	-0.008	0.75

Note:
This table compares key characteristics of ever-disclosed vs. never-disclosed (Panel A) and early-disclosed vs. late-disclosed (Panel B) states. All variables are as of 1990 except for the Democratic party vote share variable, which comes from 1988 presidential election. The last two columns show mean differences and p-values.

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Table B.4: The Effect of Discosure on Net Block Population Flow (Robustness Checks)

	Log Population			
	(1)	(2)	(3)	(4)
SFHA × Post	$-.079^{**}$	$-.301^{**}$	$-.079^{**}$	$-.080^{**}$
	(.029)	(.133)	(.031)	(.033)
Model	No Map Revisions	Time-Varying Discontinuity	Donut: 20	Donut: 40
Bandwidth	301	301	301	301
Num. obs.	710342	1915717	1763552	1607388

Note: This table is produced from equation in footnote 33. Columns (1)-(3) show results from the ever-disclosed states whereas columns (4)-(6) report results from placebo states. P.values, which are calculated using clustered standard error for columns (1)–(3) and bootstrapping for columns (2)–(6), are reported in parentheses.

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Illinois REALTORS®
RESIDENTIAL REAL PROPERTY DISCLOSURE REPORT
(765 ILCS 77/35)

NOTICE: THE PURPOSE OF THIS REPORT IS TO PROVIDE PROSPECTIVE BUYERS WITH INFORMATION ABOUT MATERIAL DEFECTS IN THE RESIDENTIAL REAL PROPERTY. THIS REPORT DOES NOT LIMIT THE PARTIES' RIGHT TO CONTRACT FOR THE SALE OF RESIDENTIAL REAL PROPERTY IN "AS IS" CONDITION. UNDER COMMON LAW, SELLERS WHO DISCLOSE MATERIAL DEFECTS MAY BE UNDER A CONTINUING OBLIGATION TO ADVISE THE PROSPECTIVE BUYERS ABOUT THE CONDITION OF THE RESIDENTIAL REAL PROPERTY EVEN AFTER THE REPORT IS DELIVERED TO THE PROSPECTIVE BUYER. COMPLETION OF THIS REPORT BY THE SELLER CREATES LEGAL OBLIGATIONS ON THE SELLER; THEREFORE SELLER MAY WISH TO CONSULT AN ATTORNEY PRIOR TO COMPLETION OF THIS REPORT.

Property Address: ______________________________

City, State & Zip Code: ______________________________

Seller's Name: ______________________________

This Report is a disclosure of certain conditions of the residential real property listed above in compliance with the Residential Real Property Disclosure Act. This information is provided as of ______________________________, 20___, and does not reflect any changes made or occurring after that date or information that becomes known to the seller after that date. The disclosures herein shall not be deemed warranties of any kind by the seller or any person representing any party in this transaction.

In this form, "am aware" means to have actual notice or actual knowledge without any specific investigation or inquiry. In this form, a "material defect" means a condition that would have a substantial adverse effect on the value of the residential real property or that would significantly impair the health or safety of future occupants of the residential real property unless the seller reasonably believes that the condition has been corrected.

The seller discloses the following information with the knowledge that even though the statements herein are not deemed to be warranties, prospective buyers may choose to rely on this information in deciding whether or not and on what terms to purchase the residential real property.

The seller represents that to the best of his or her actual knowledge, the following statements have been accurately noted as "yes" (correct), "no" (incorrect), or "not applicable" to the property being sold. If the seller indicates that the response to any statement, except number 1, is yes or not applicable, the seller shall provide an explanation, in the additional information area of this form.

	YES	NO	N/A	
1.	___	___	___	Seller has occupied the property within the last 12 months. (No explanation is needed.)
2.	___	___	___	I am aware of flooding or recurring leakage problems in the crawl space or basement.
3.	___	___	___	I am aware that the property is located in a flood plain or that I currently have flood hazard insurance on the property.
4.	___	___	___	I am aware of material defects in the basement or foundation (including cracks and bulges).
5.	___	___	___	I am aware of leaks or material defects in the roof, ceilings, or chimney.
6.	___	___	___	I am aware of material defects in the walls, windows, doors, or floors.
7.	___	___	___	I am aware of material defects in the electrical system.
8.	___	___	___	I am aware of material defects in the plumbing system (includes such things as water heater, sump pump, water treatment system, sprinkler system, and swimming pool).
9.	___	___	___	I am aware of material defects in the well or well equipment.
10.	___	___	___	I am aware of unsafe conditions in the drinking water.
11.	___	___	___	I am aware of material defects in the heating, air conditioning, or ventilating systems.
12.	___	___	___	I am aware of material defects in the fireplace or wood burning stove.
13.	___	___	___	I am aware of material defects in the septic, sanitary sewer, or other disposal system.
14.	___	___	___	I am aware of unsafe concentrations of radon on the premises.
15.	___	___	___	I am aware of unsafe concentrations of or unsafe conditions relating to asbestos on the premises.
16.	___	___	___	I am aware of unsafe concentrations of or unsafe conditions relating to lead paint, lead water pipes, lead plumbing pipes or lead in the soil on the premises.
17.	___	___	___	I am aware of mine subsidence, underground pits, settlement, sliding, upheaval, or other earth stability defects on the premises.
18.	___	___	___	I am aware of current infestations of termites or other wood boring insects.
19.	___	___	___	I am aware of a structural defect caused by previous infestations of termites or other wood boring insects.
20.	___	___	___	I am aware of underground fuel storage tanks on the property.
21.	___	___	___	I am aware of boundary or lot line disputes.
22.	___	___	___	I have received notice of violation of local, state or federal laws or regulations relating to this property, which violation has not been corrected.
23.	___	___	___	I am aware that this property has been used for the manufacture of methamphetamine as defined in Section 10 of the Methamphetamine Control and Community Protection Act.

Note: These disclosures are not intended to cover the common elements of a condominium, but only the actual residential real property including limited common elements allocated to the exclusive use thereof that form an integral part of the condominium unit.

Note: These disclosures are intended to reflect the current condition of the premises and do not include previous problems, if any, that the seller reasonably believes have been corrected.

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Figure B.1: Example of the Home Seller Disclosure Form (IL)

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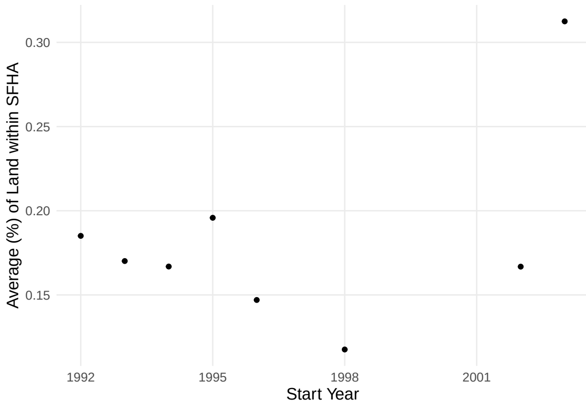
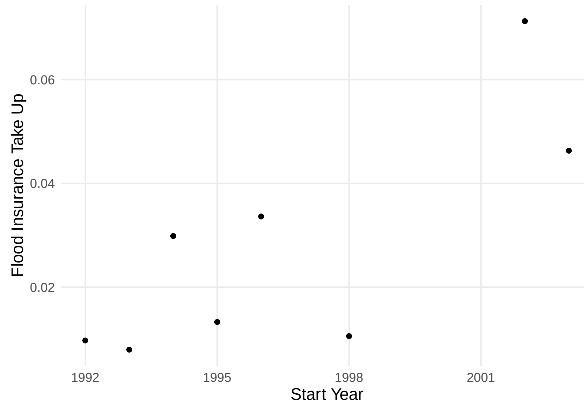
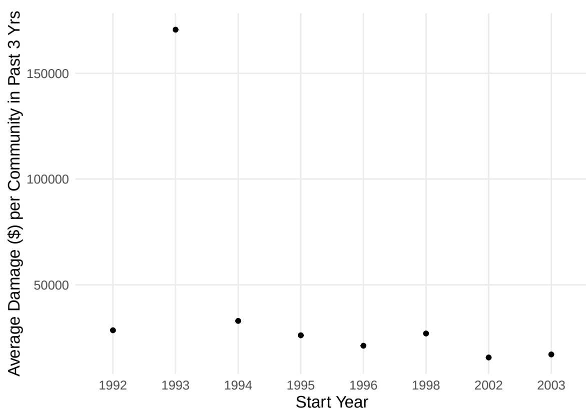


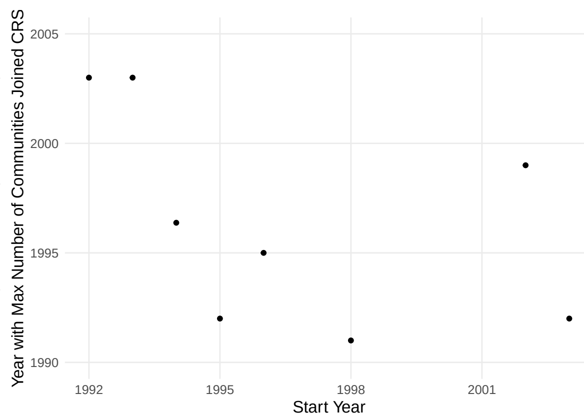
(a) Timing and (%) of Land in SFHA



(b) Timing and Flood Insurance Take Up



(c) Timing and Flood Insurance Claims



(d) Timing and Participation to the CRS Program

Figure B.2: Correlation Between Disclosure Timing and Flood Profiles. These figures plot the disclosure policy timing against (a) share of SFHA area, (b) flood insurance take up in the past three years, (c) flood insurance claims amount in the past three years, and (d) flood policy (timing of participation to the Community Ratings System). Values on the y-axis is pooled across all states with the same treatment or event year.

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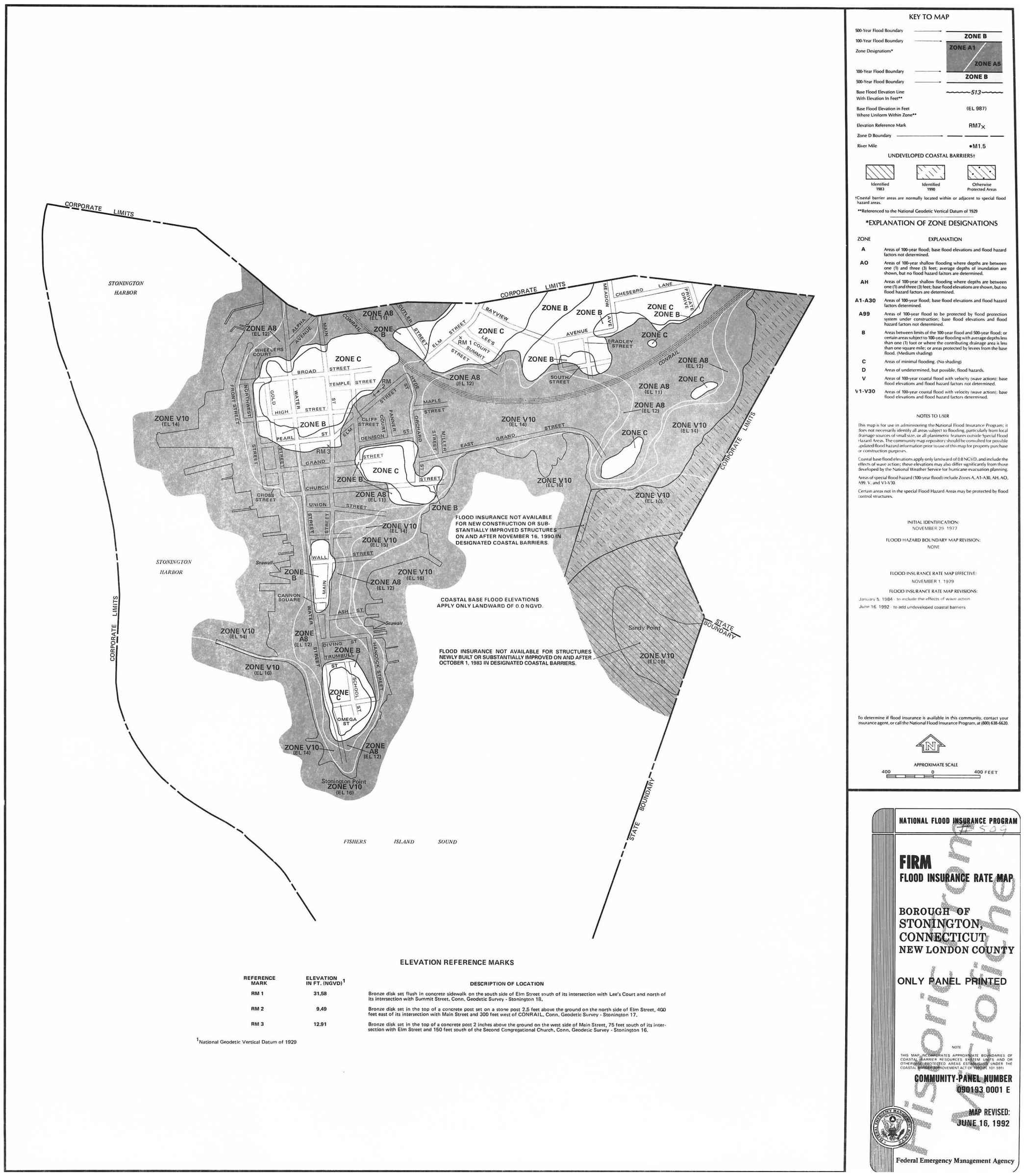

Figure B.3: Sample Flood Insurance Rate Map (Borough of Stonington, CT)

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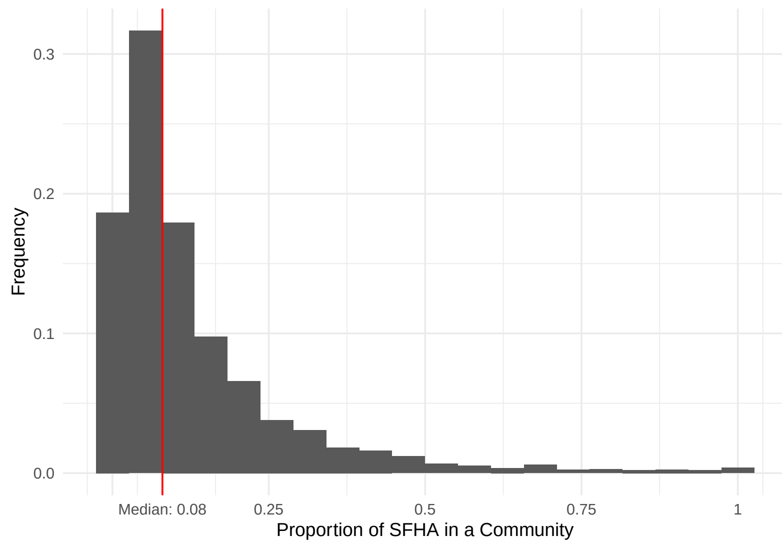

Figure B.4: Histogram of the Proportion of the SFHA at the Community Level. The plot shows the distribution of the SFHA ratio for the 8,175 communities that are on the Q3 map (first generation of digitized flood map) and in the 26 ever-disclosed states.

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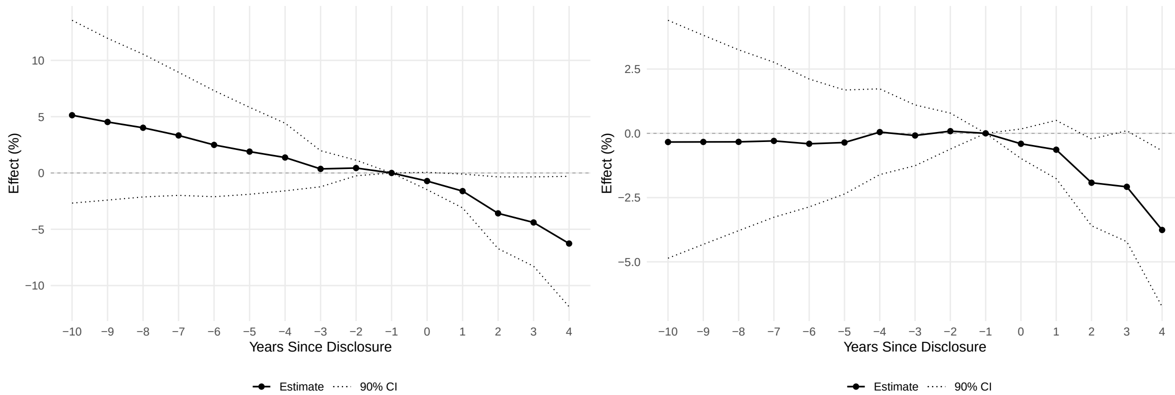

(a) Inc. Never-Adopted States (b) Exc. Map Updates

Figure B.5: The Effect of Disclosure Policy on Community Population

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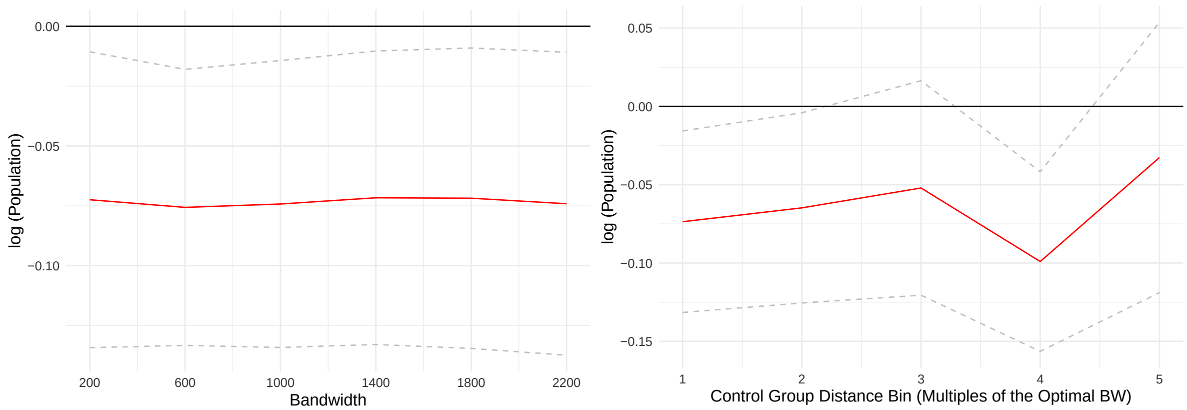

(a) Effect by Bandwidth Size

(b) Effect by Varying Control Units

Figure B.6: The Effect of the Disclosure Requirement on Population by Bandwidths and Control Group Distance Bin. Panel (a) plots $\hat{\delta_6}$ from equation (2) with log of population as an outcome variable for a range of bandwidths. Panel (b) plots $\hat{\delta_6}$ from equation (2) for control groups of varying distance. Note, the horizontal axis of Panel (b) indicates the distance bin of control group in multiples of optimal bandwidth for log of population variable (i.e., distance bin r on x-axis indicates that control group blocks are within $(r-1)$ and r times optimal bandwidth). In all panels, the level of observation is census block, which is the smallest census geographical unit. Standard errors are clustered at the state level.

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